Università di Genova

MeshBoat

Secure your floating home

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Outline

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02 Proposed Solution

03 Architecture

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03.2 Software

04 Visualization & Interaction

05 Conclusion

Problem

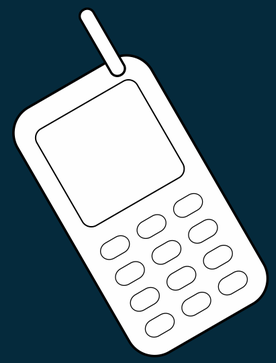
+14%

Maritime crimes compared to the previous year

Looting focusing on high-value, easily transportable assets

131

Maritime crimes in Italy in 2025

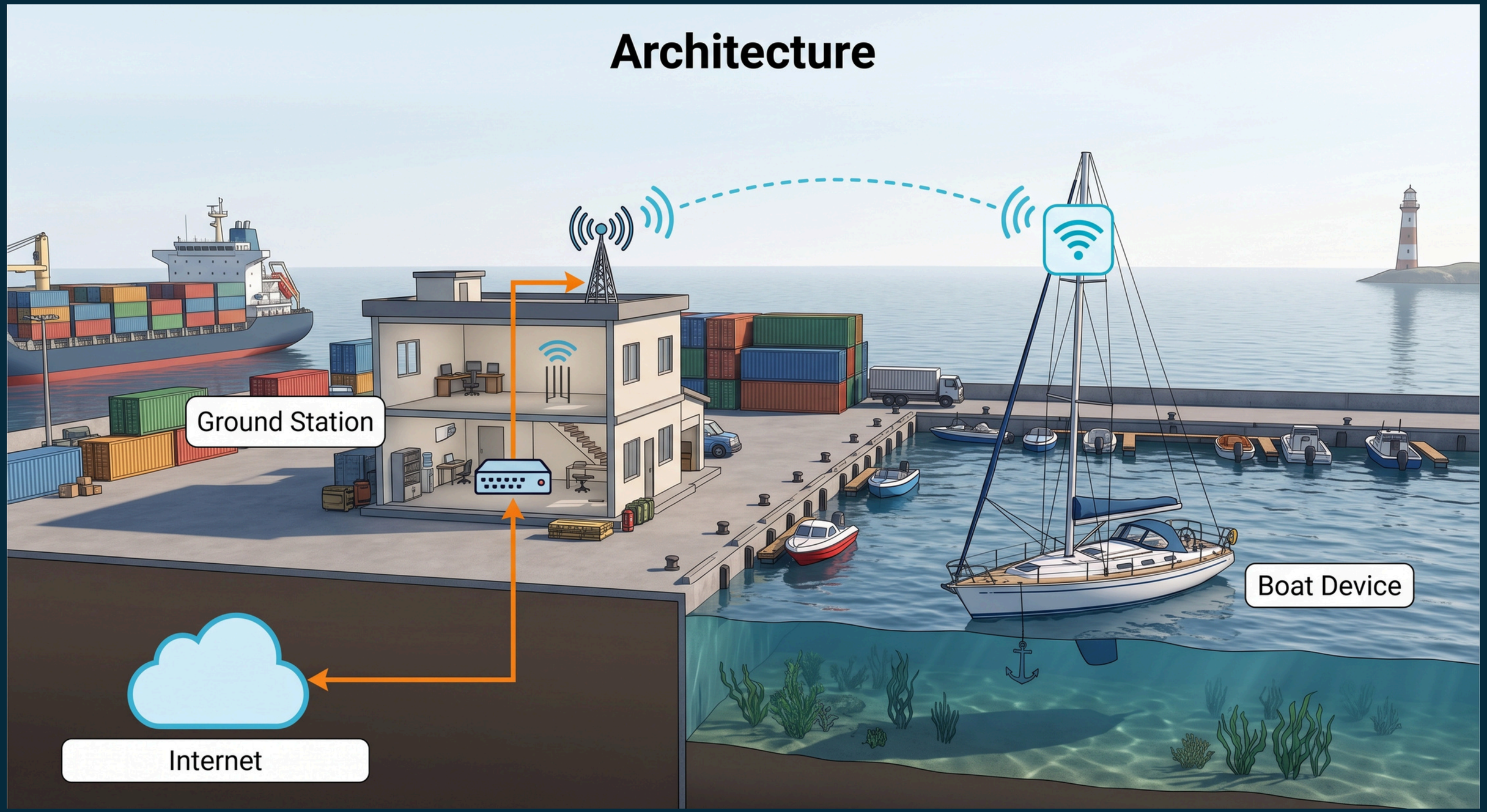


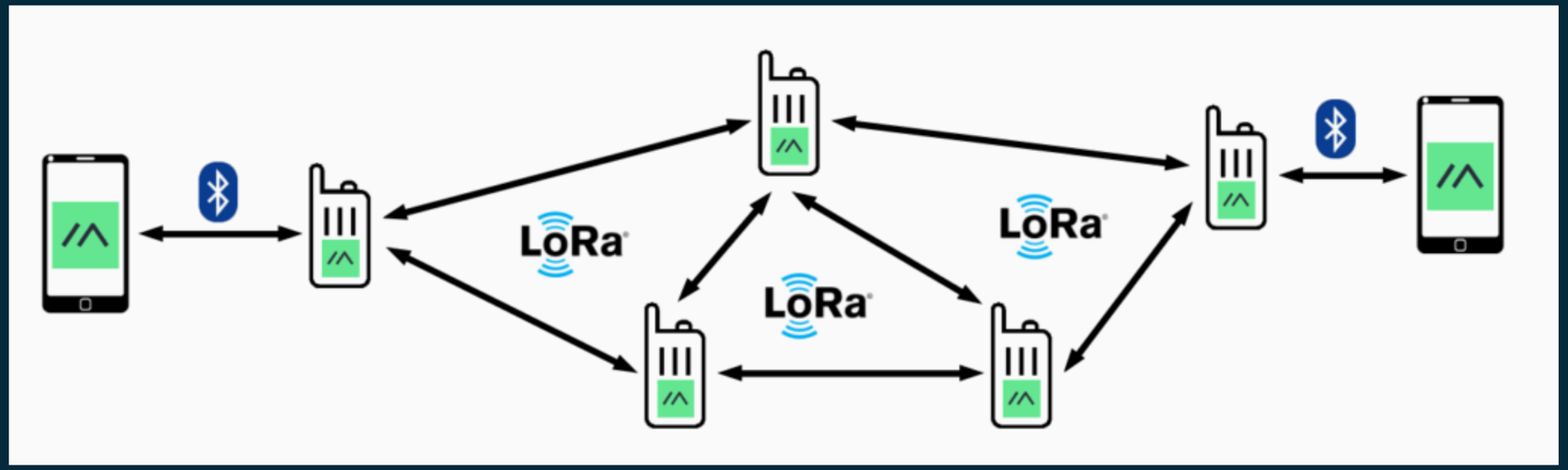
?

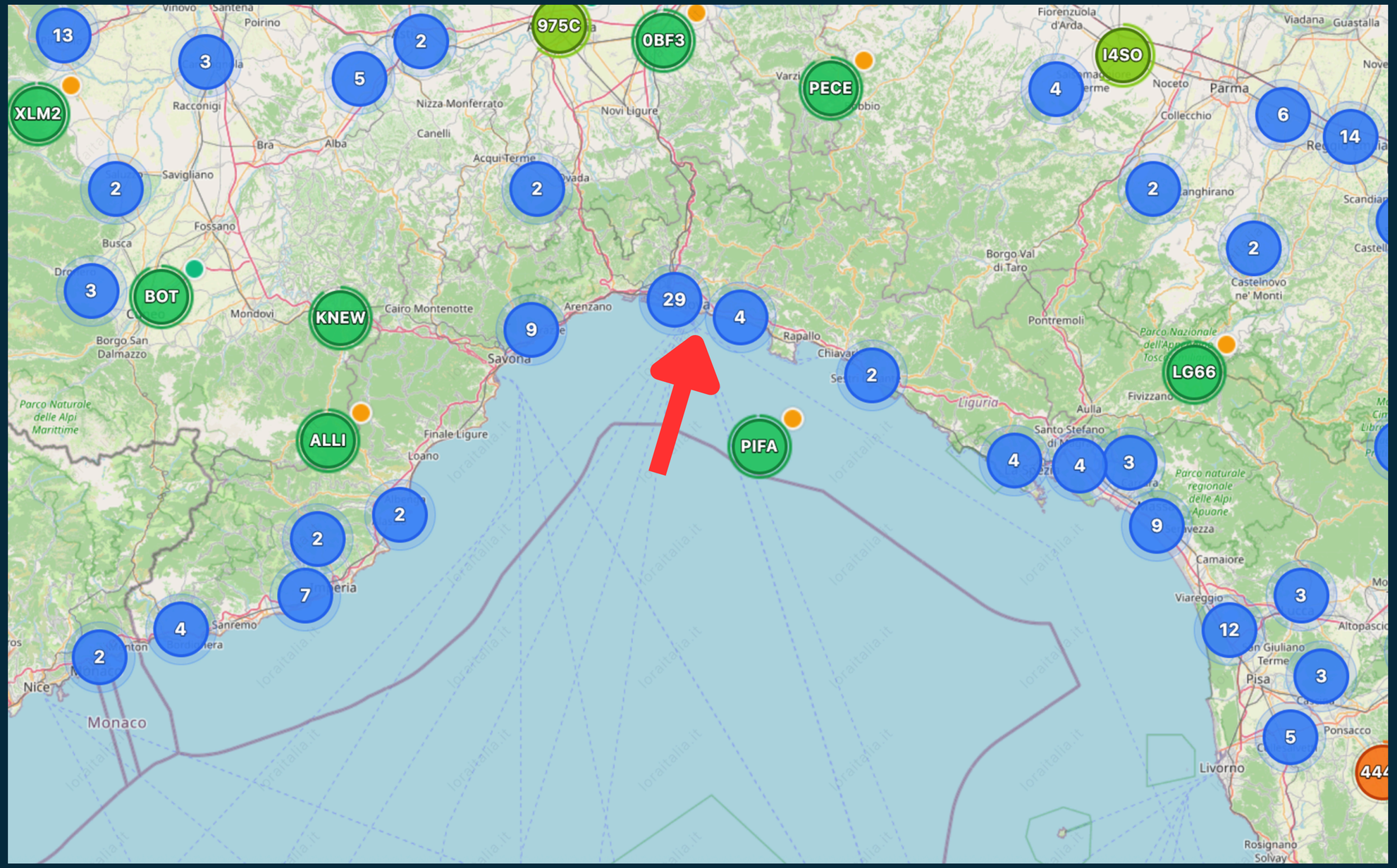
MeshBoat
Secure your floating home

Proposed Solution

Architecture




LoRa
LoRa
LoRa



loraitalia.it

Network Analysis Summary

Comprehensive analysis of topology, centrality, and connectivity metrics.

MQTT BROKER

EXPORT PDF

TOTAL NODES	TOTAL EDGES	NETWORK DIAMETER	AVERAGE PATH LENGTH
1578	1833	15 Max hops	5.55 Avg hops

NETWORK ASSORTATIVITY

-0.028

-1.0 DISASSORTATIVE

0.0 NEUTRAL

+1.0 ASSORTATIVE

Disassortative Network: High degree nodes are less connected to one another and the failure of a high degree node would have more impact on the connectedness of the network.

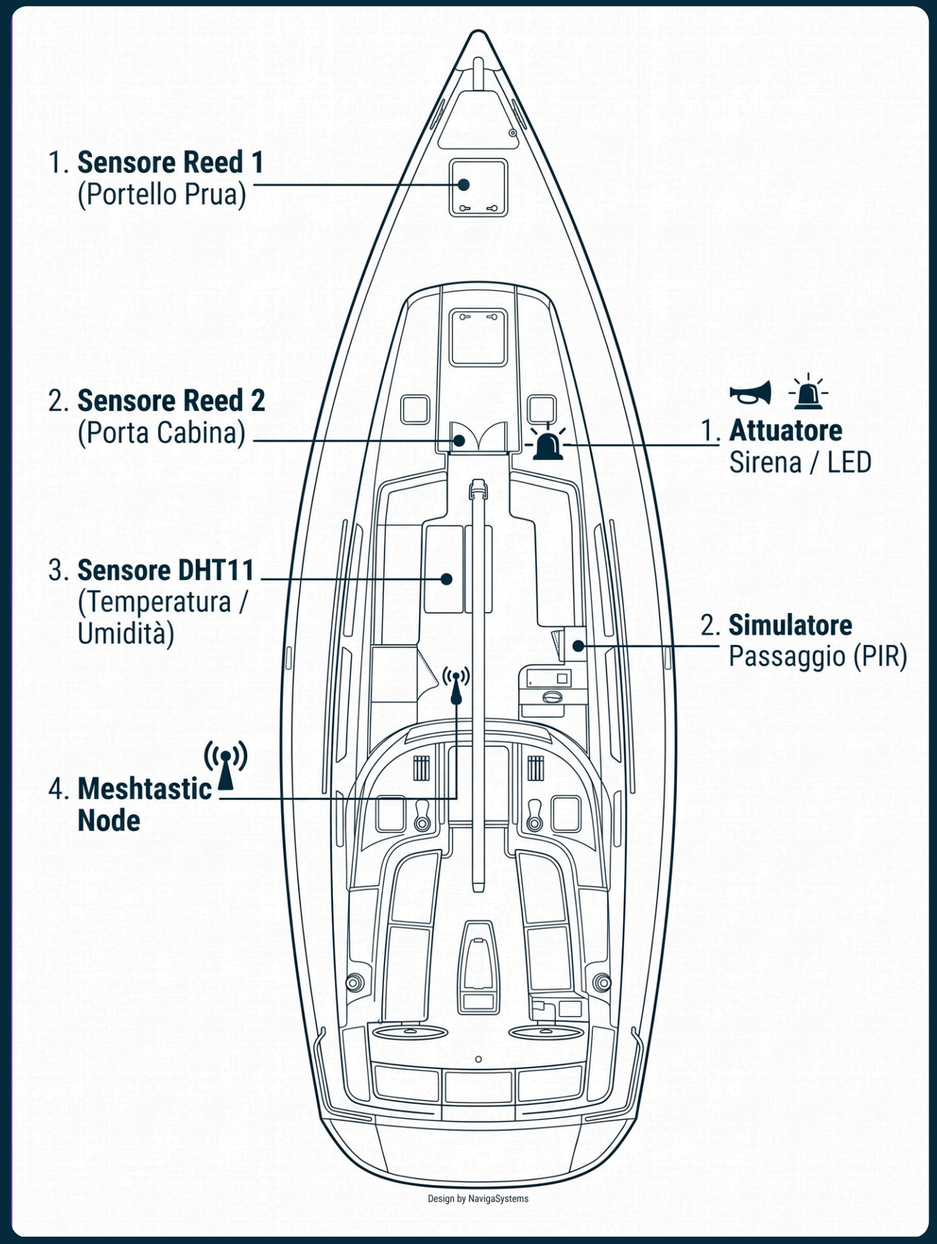
meshintel.it

Our Vision

A network of boats

Meshtastic in nautic

By connecting boats through a LoRa mesh network, we eliminate the need of traditional cellular coverage. This ensures that critical security alerts are always transmitted, fostering a safer and more connected nautical community.



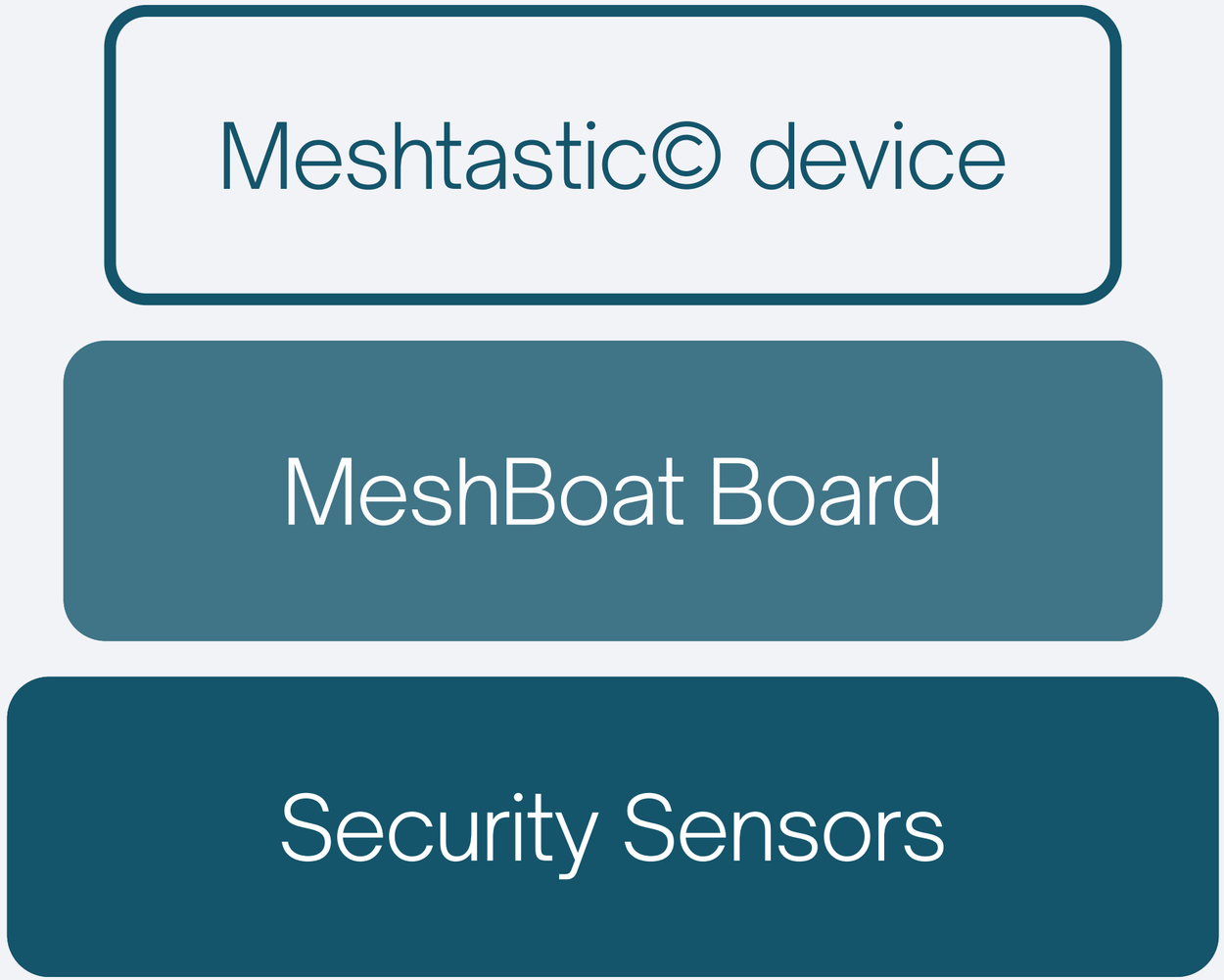
Security

- **Comprehensive Sensor Array:** Magnetic Reed switches (doors/hatches) and motion sensor that secure both perimeter and interior.
- **Environmental Monitoring:** A DHT11 sensor to track temperature and humidity levels.
- **Instant Local Deterrent:** A high-intensity siren and LED trigger immediately upon intrusion.
- **Real-Time LoRa Alerts:** The onboard Meshtastic node broadcasts alerts instantly, working without cellular coverage.

Architecture

Boat Device

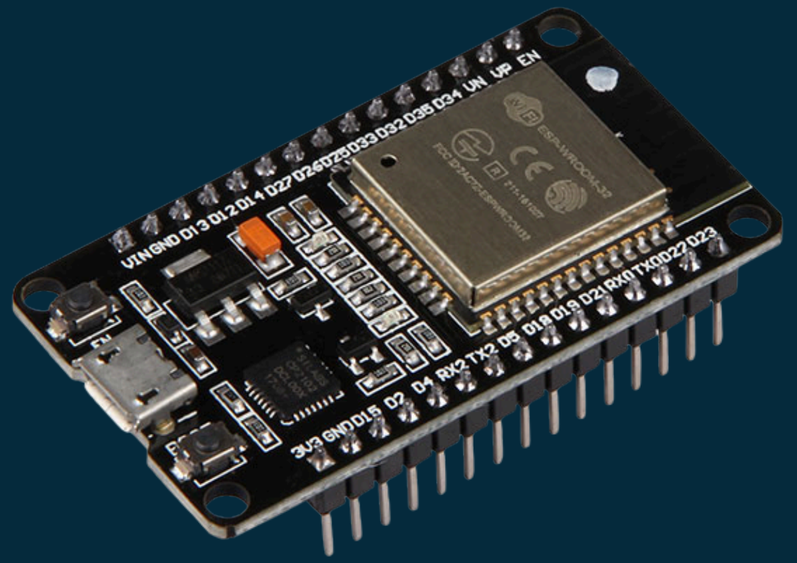
AI generated



solar powered

Meshtastic© device

MeshBoat Board



XIAO ESP32 powered

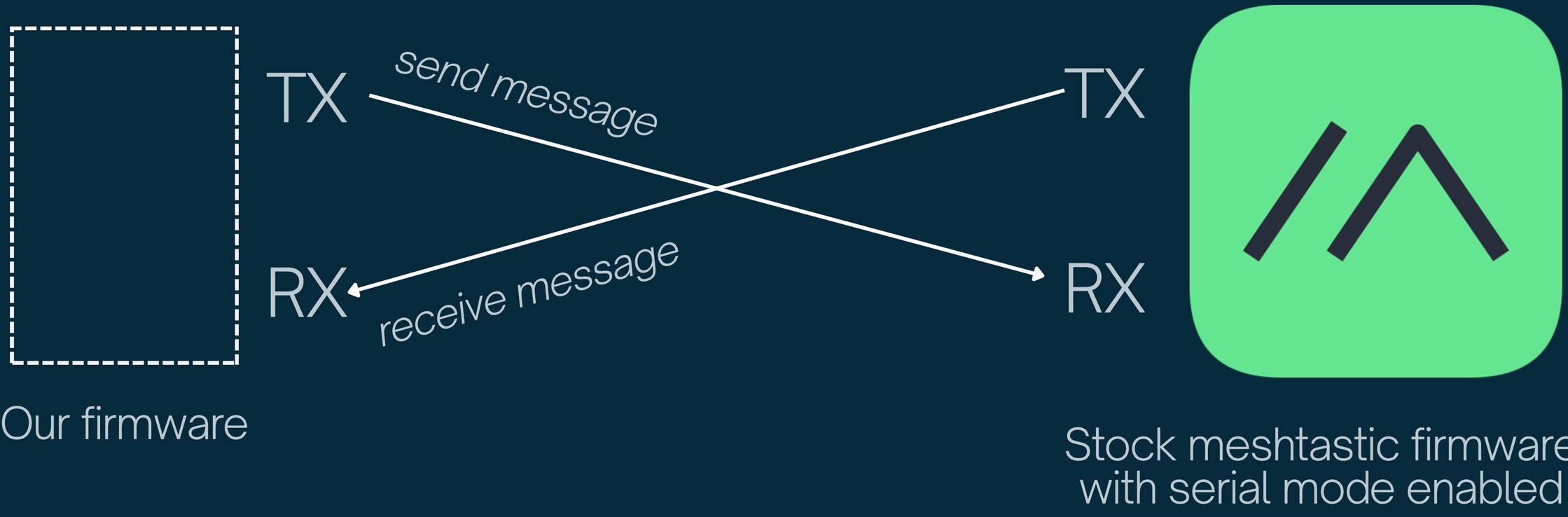
ESP32-wroom-32 powered

Why two MCUs?

Customizing Meshtastic firmware is a really **bad idea** !

- Breaking the mesh
- Isolation
- Updates
- Merging code
- Testing
- Mobile app
- ...

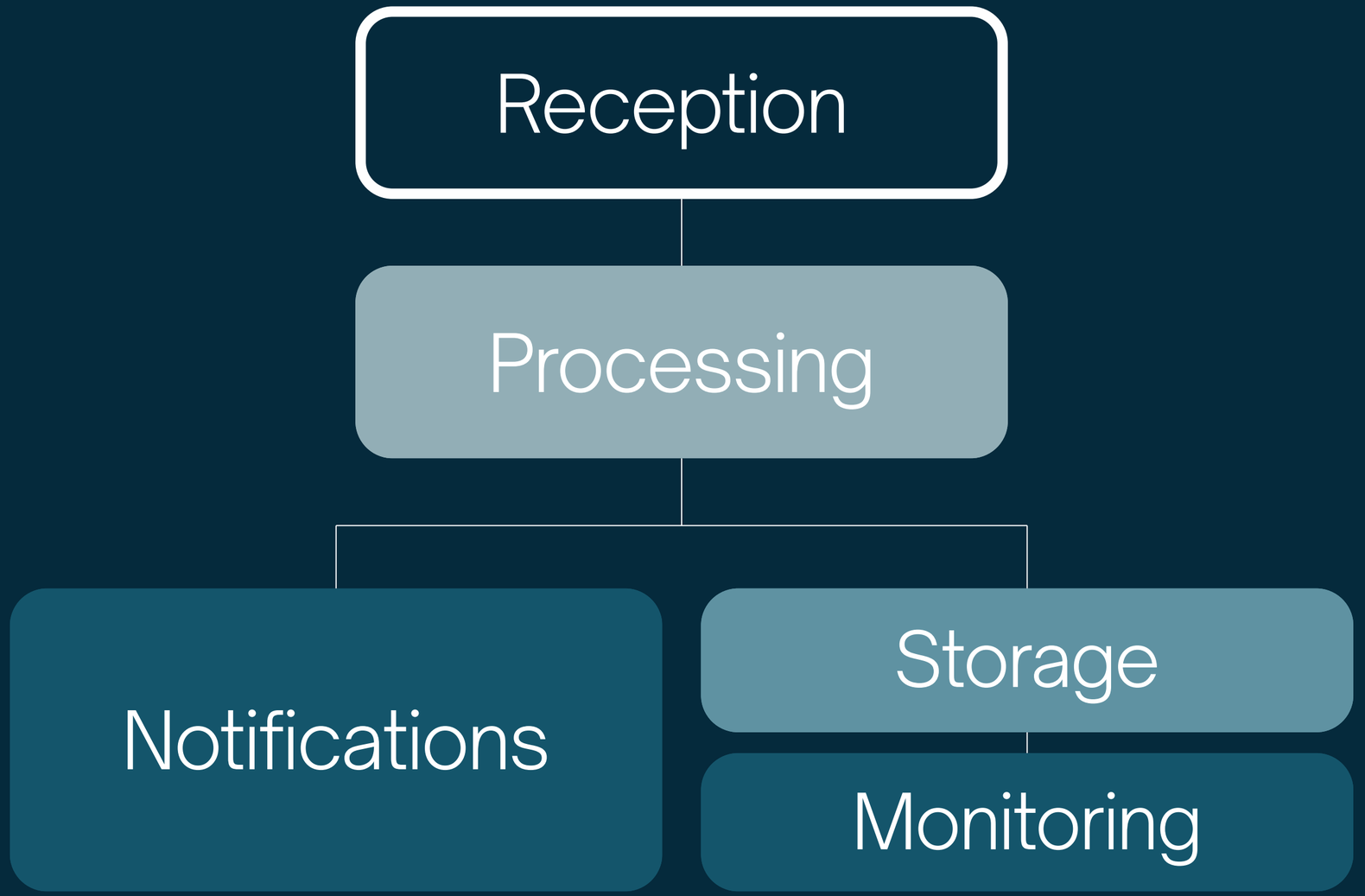
Just use UART !



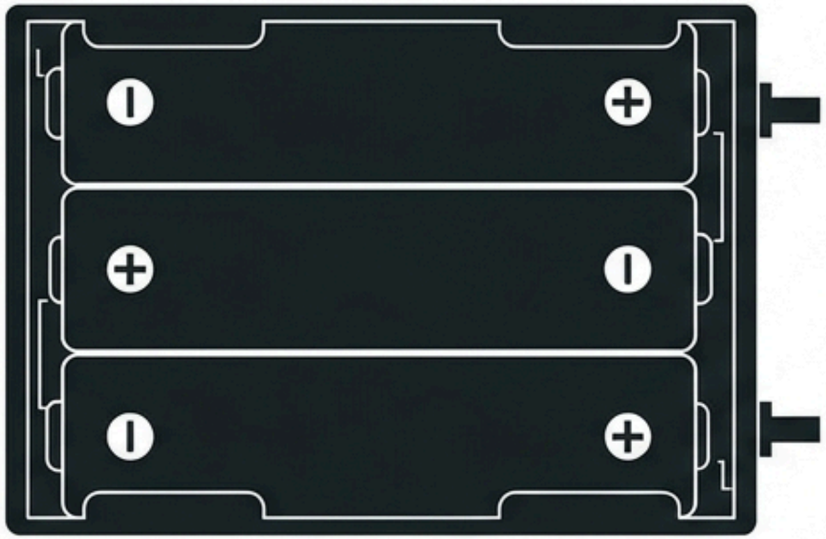
**All messages are sent on a private channel to avoid spam*

Ground Station

AI generated

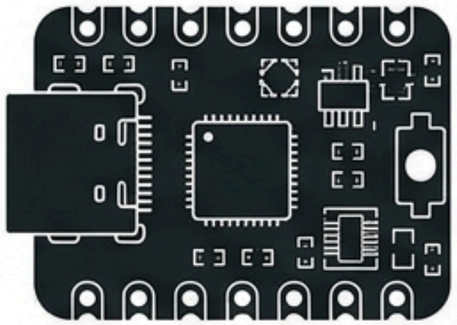

Reception
Processing
Notifications
Storage
Monitoring

Hardware Details & Tinkering

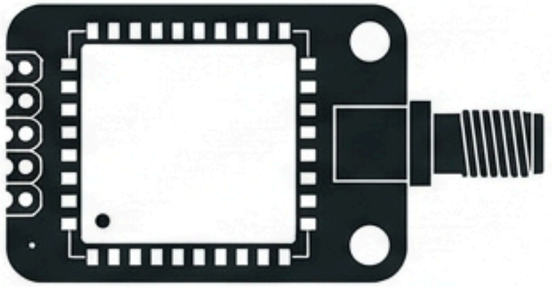


18650

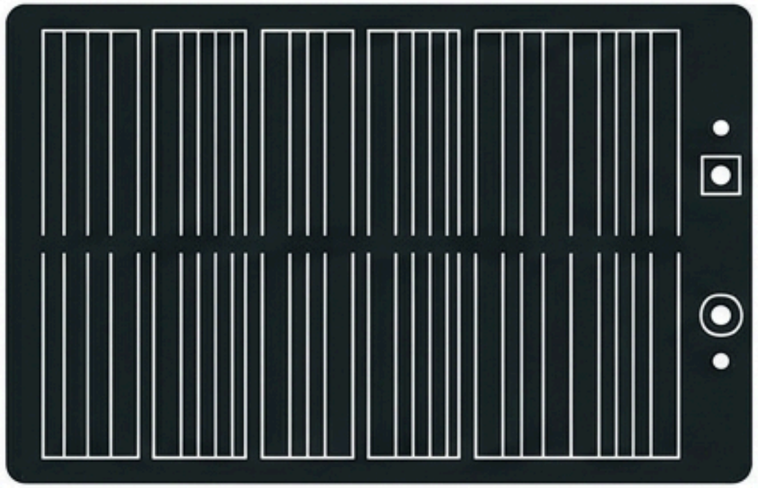
18650



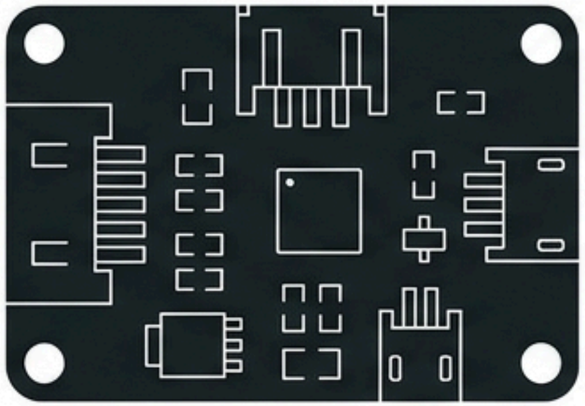
XIAO ESP32S3



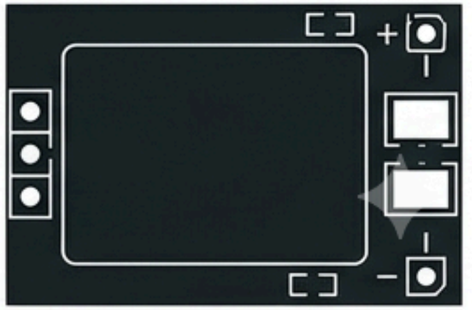
Wio-SX1262



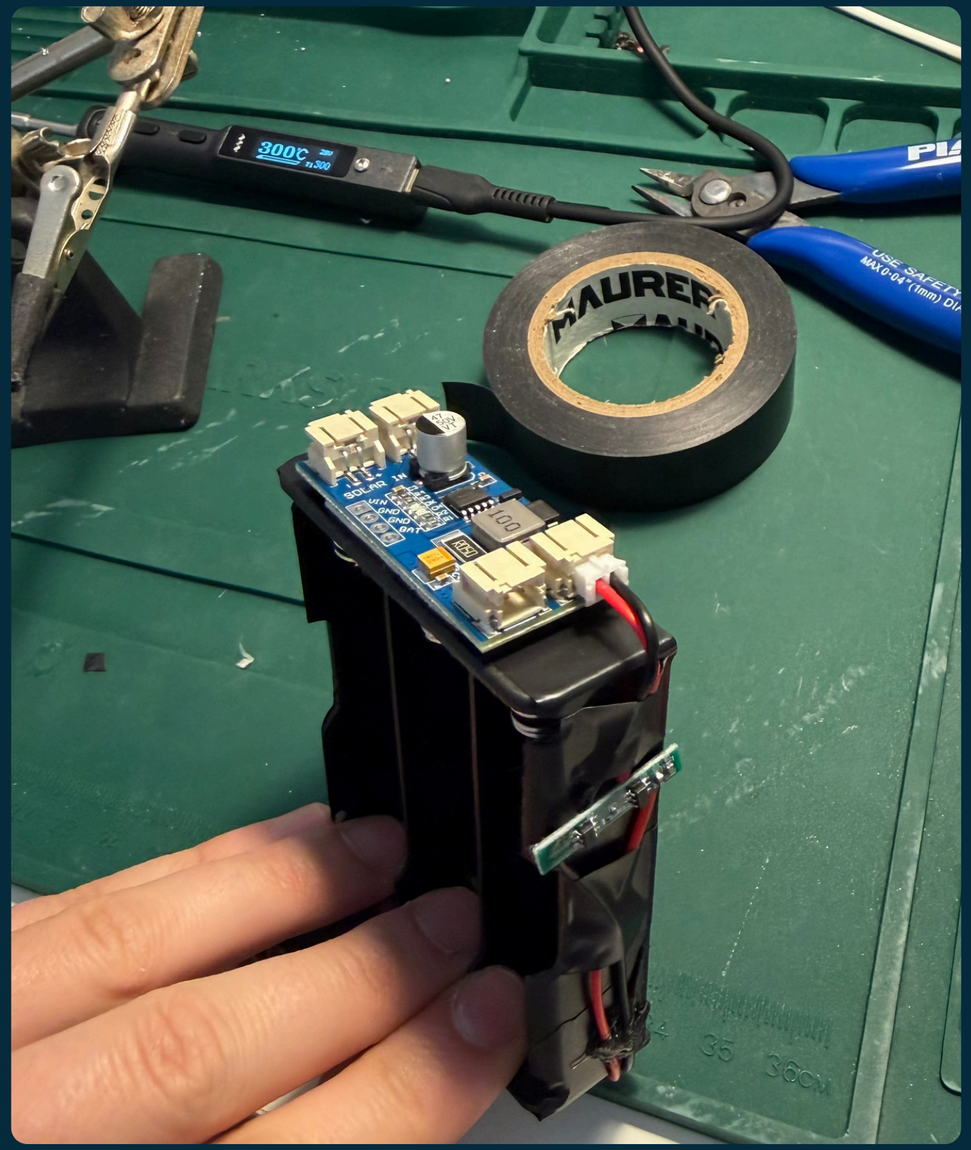
5W 6V solar panel



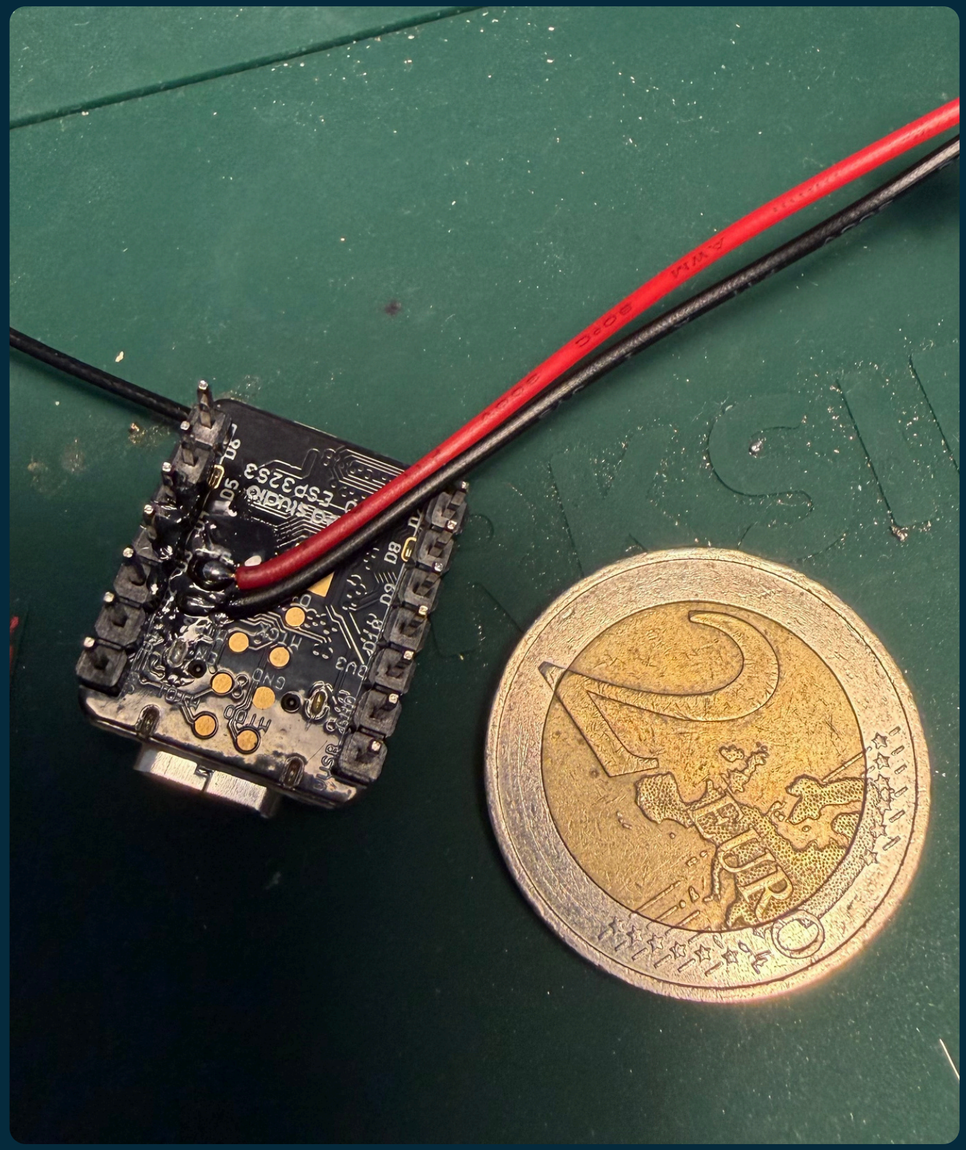
CN3791 solar charging board



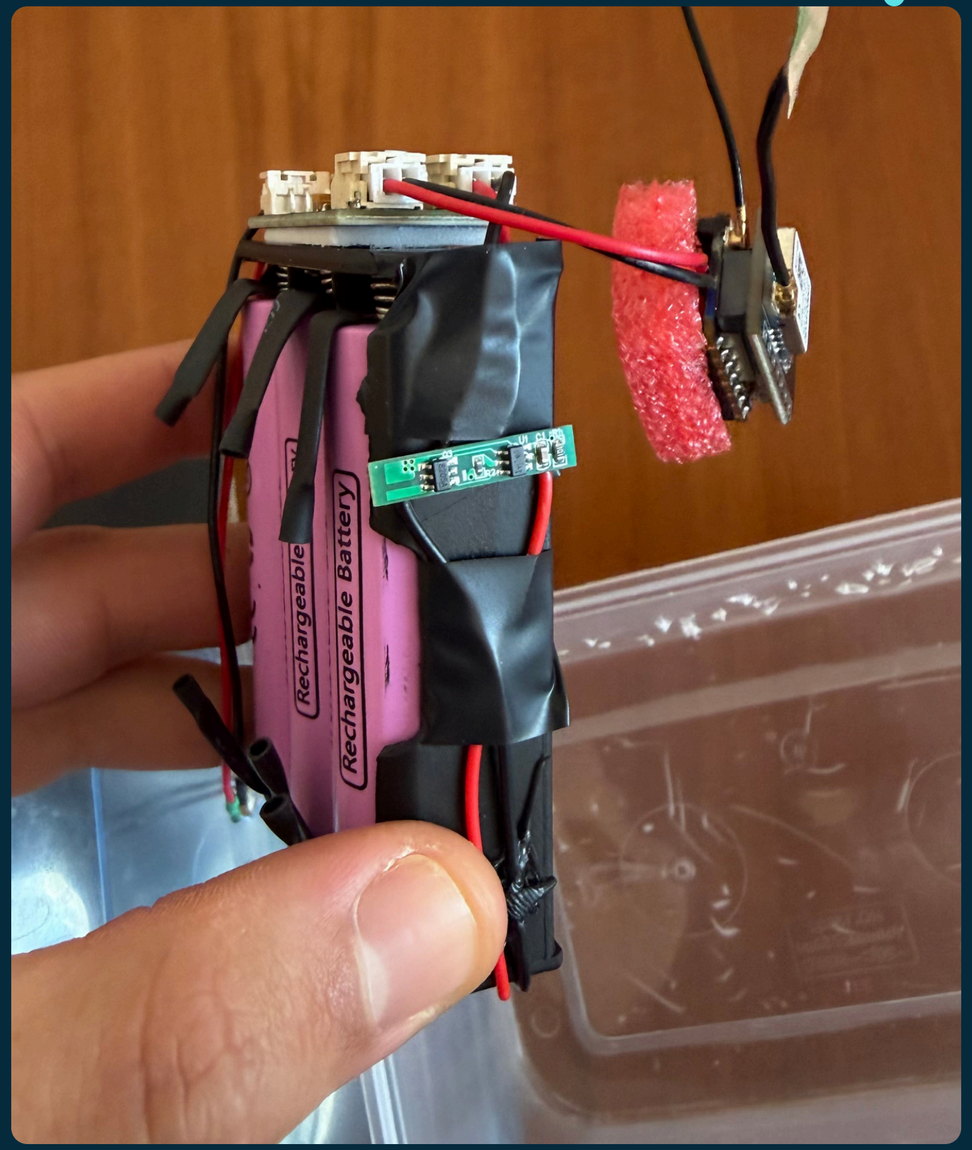
Battery BMS



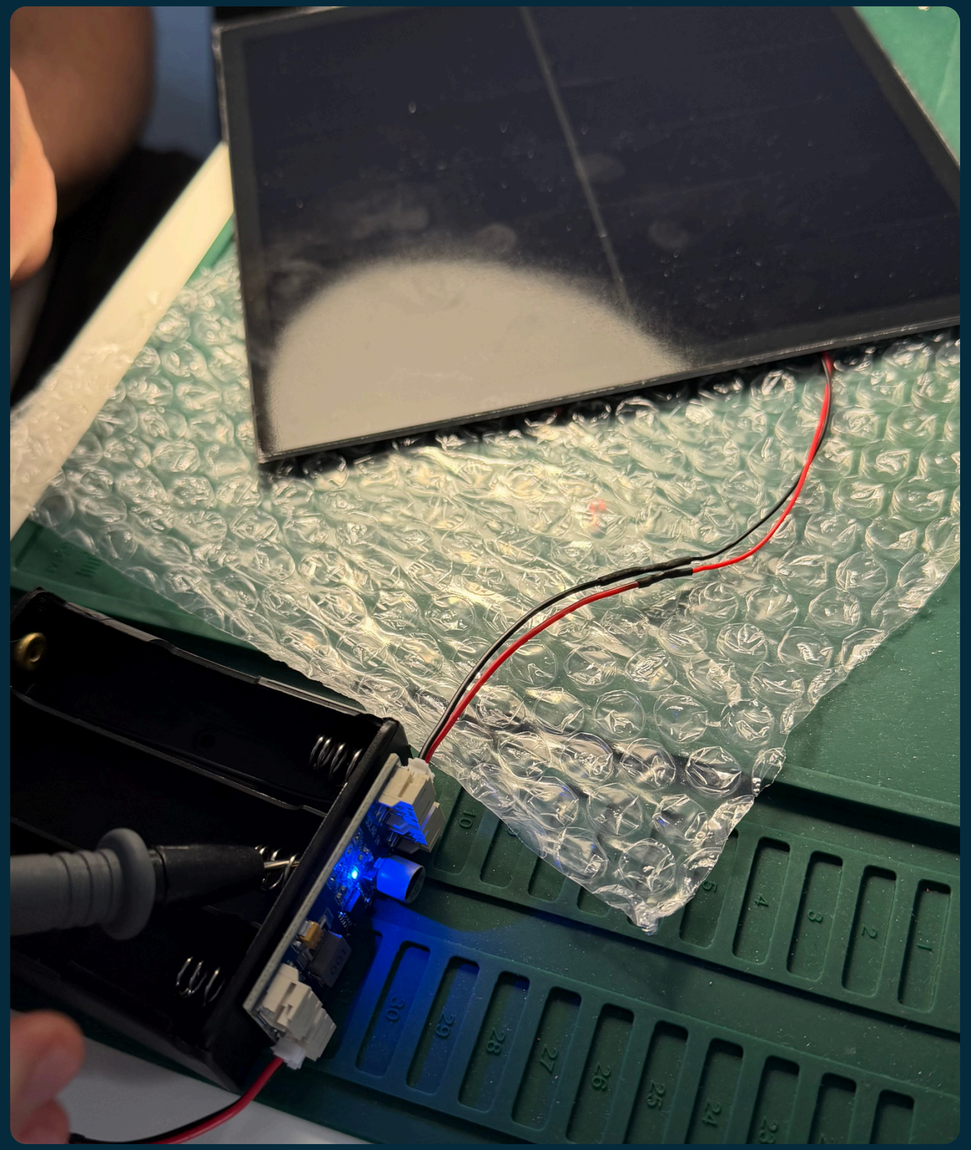
3x18650 battery pack with protection circuit



Soldering XIAO ESP32 battery pads

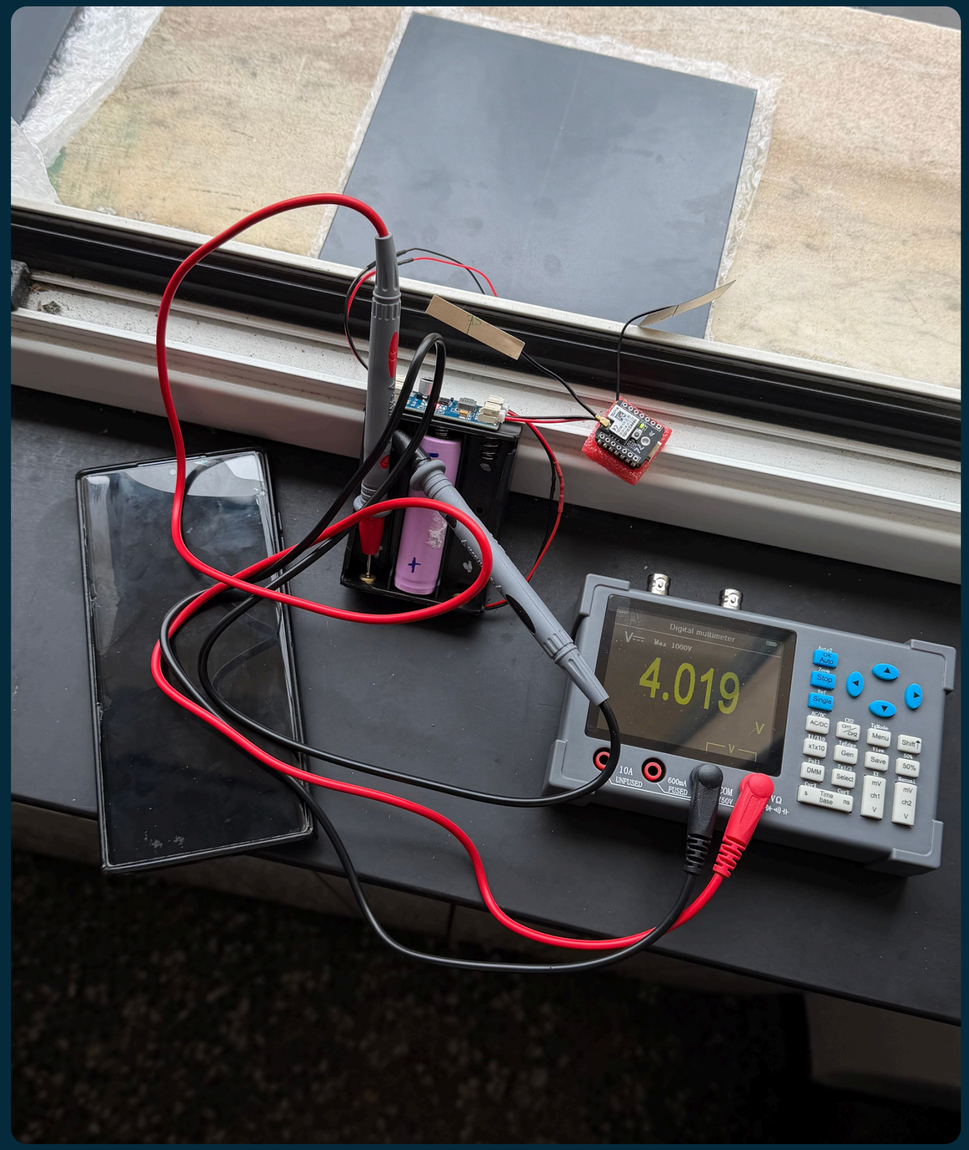


Battery powered node assembled

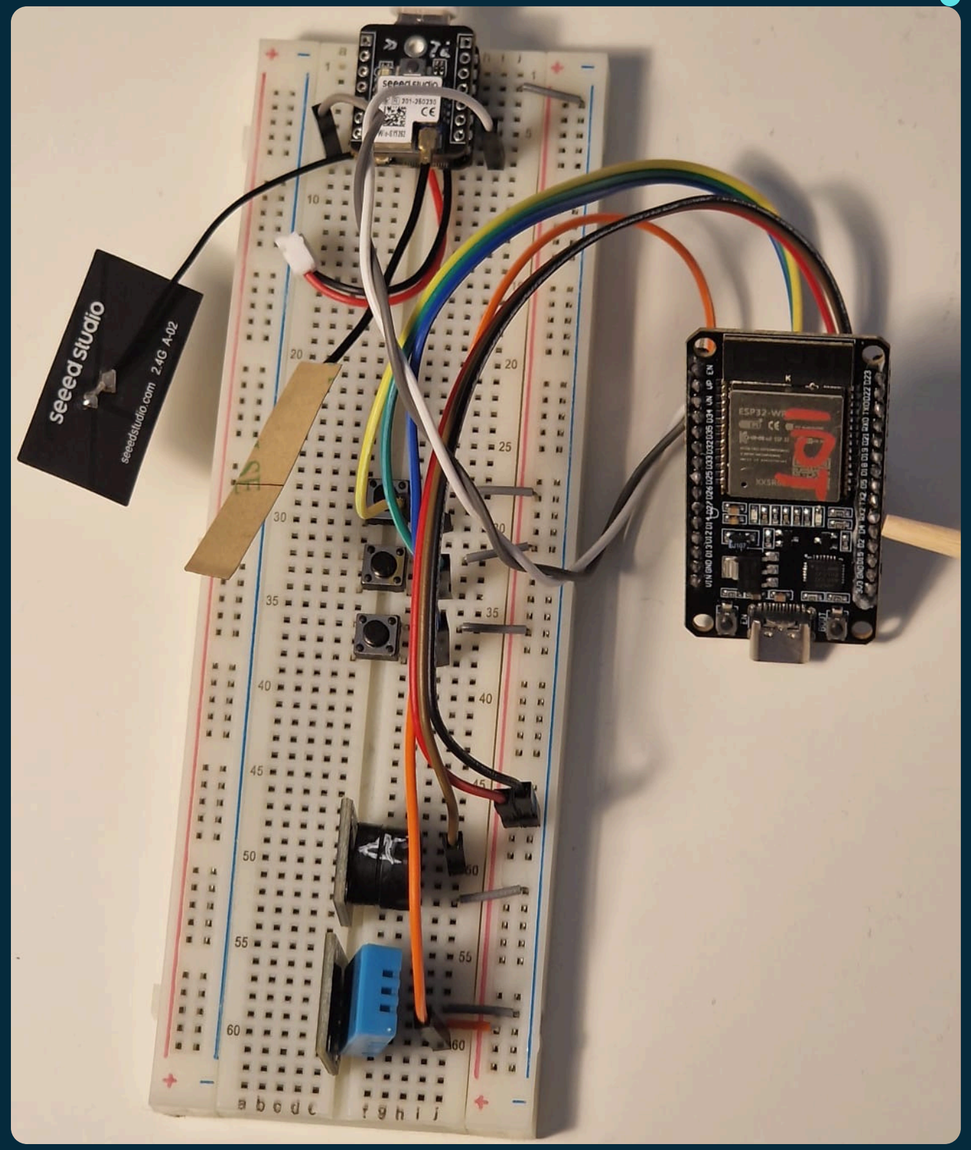


6V Solar Panel with (fake) MPPT circuit

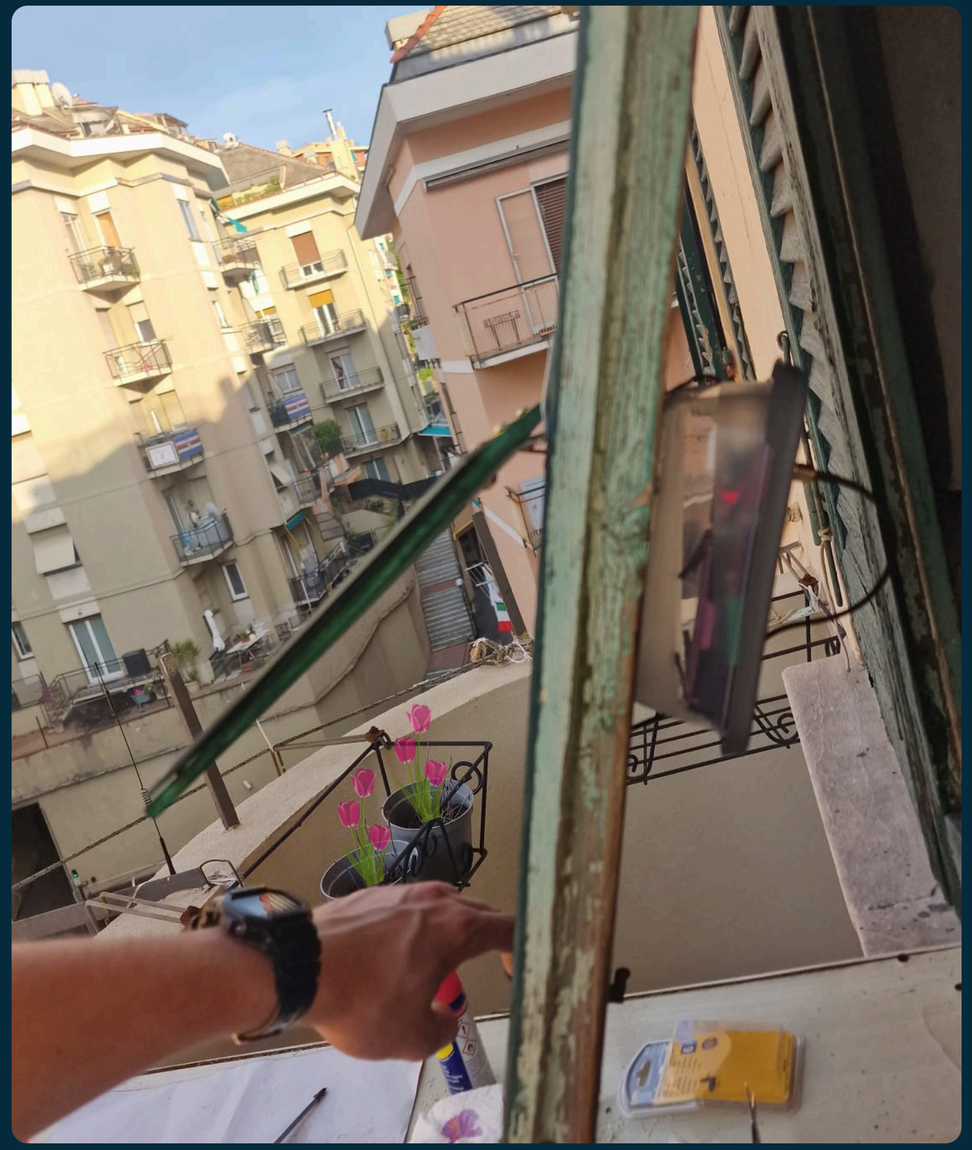
Learning a new skill



Succesfully charging a battery using the sun

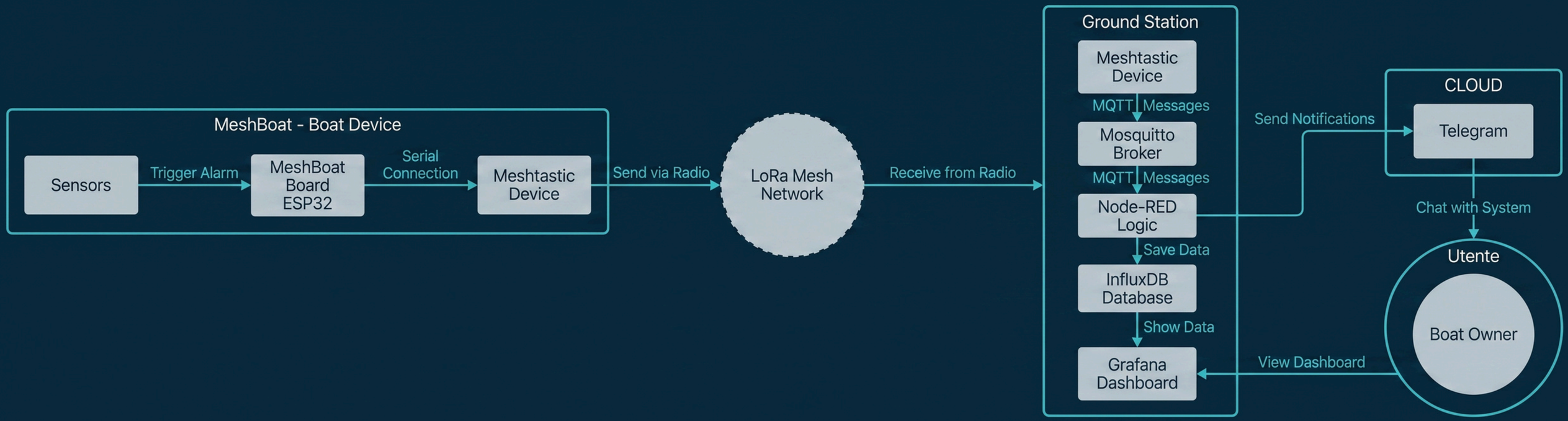


MeshBoat Board prototype



Deployed “Ground Station”

Hardware and Software togeter...



Software

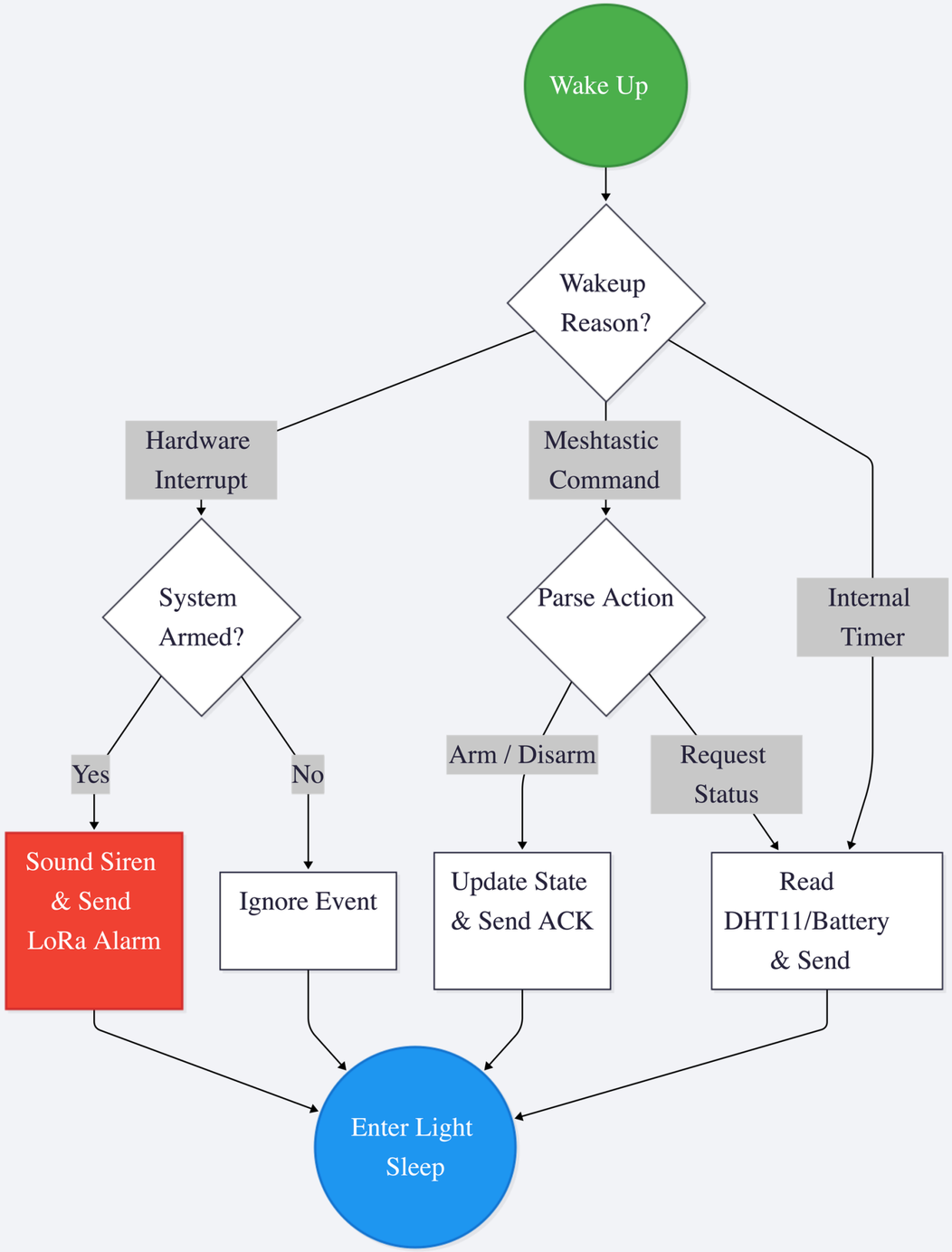
Pull-up
Pull-down

Reed door/hatch sensors & PIR motion sensor rest at 0, trigger to 1.
The Meshtastic radio's UART line rests at 1 (serial idle standard), drops to 0 when a message starts.

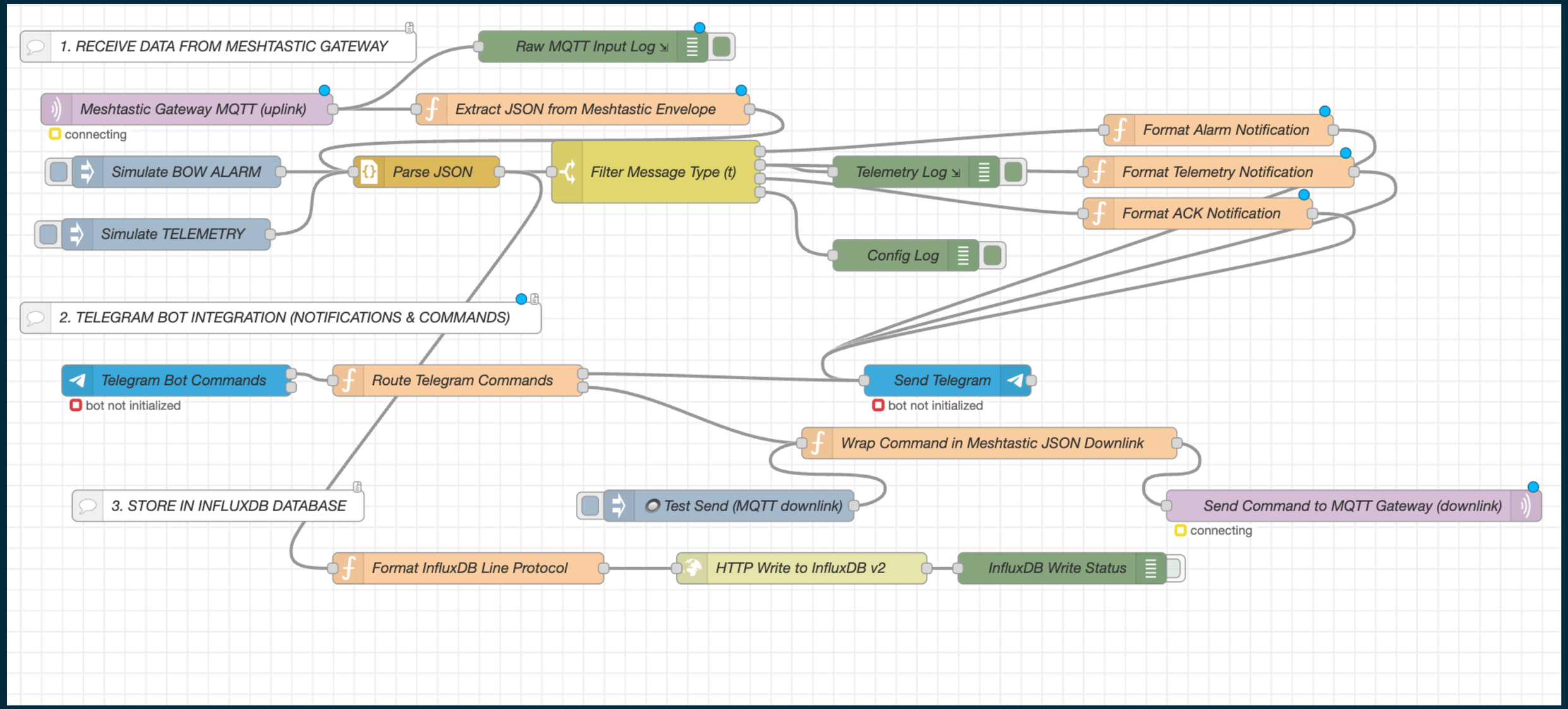
Interrupt

EXT1 wakes the ESP32 when a physical sensor triggers (hatch, cabin door, PIR).
EXT0 wakes it when a remote command arrives over radio (UART RX goes LOW).

Arduino code schema



Nodered Flow



Visualization & Interaction

Telegram Bot

Stay Connected, Anywhere

Control your boat's security system directly from your smartphone using our dedicated Telegram Bot.
Available Commands:

- **/arm** – Instantly activates the onboard security perimeter and sensors.
- **/disarm** – Deactivates the alarm system to allow safe entry.
- **/status** – Requests real-time telemetry (temperature, humidity, battery) and the current system state.

Dashboard

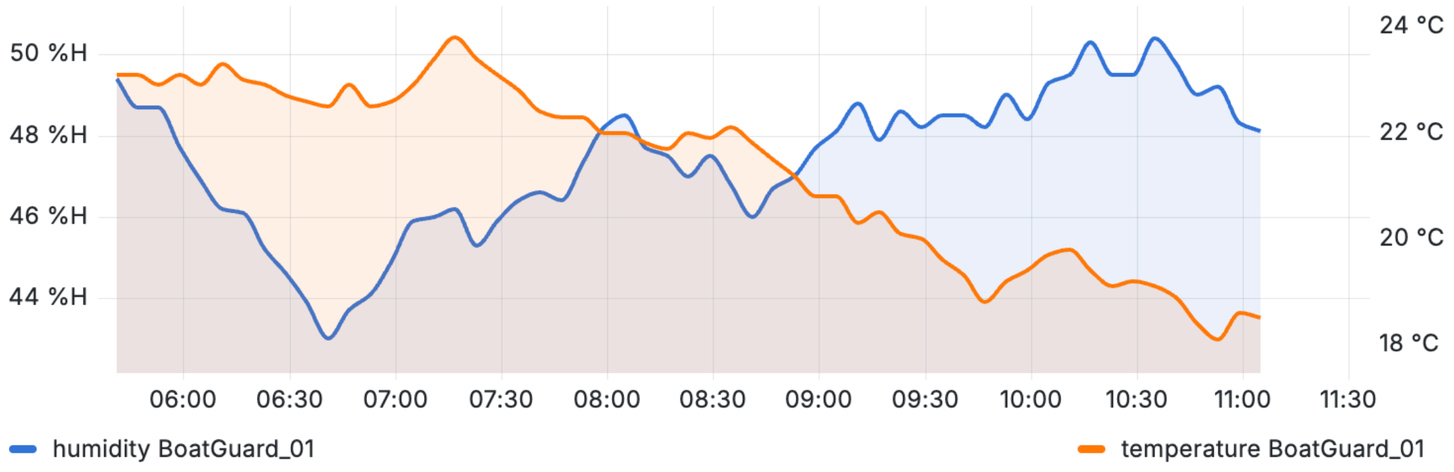
System State (Armed State)

ARMED

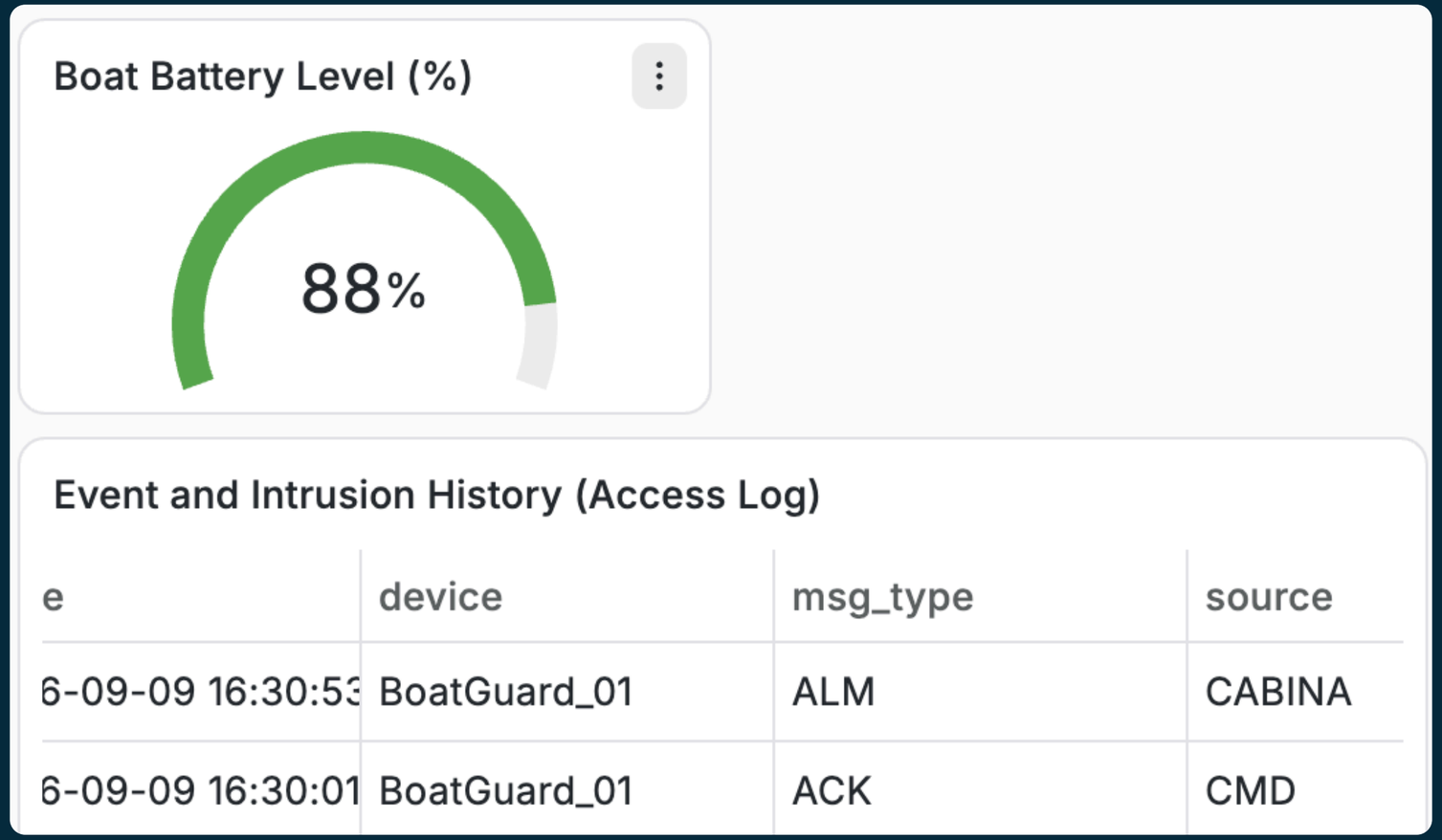
Intrusion / Alarm State

NORMAL (NO ALARMS)

Environmental Telemetry Chart (DHT11 - Temp & Humidity)



Dashboard


Boat Battery Level (%)
88%
Event and Intrusion History (Access Log)
e
device
msg_type
source
6-09-09 16:30:53
BoatGuard_01
ALM
CABINA
6-09-09 16:30:01
BoatGuard_01
ACK
CMD

Conclusion

Improvements & Scalability

MeshBoatGuard is a low-budget prototype demonstrating the viability of a LoRa mesh network for maritime security, with a fully scalable architecture. Future upgrades:

- **Real sensors** instead of prototype push-buttons
- **Real battery telemetry**
- **High-gain marine antenna** for extended range
- **More powerful solar panel** for 24/7 autonomy even in cloudy weather

Thank You !